Process Characterization Plan

A-Mab Drug Substance — Anion Exchange Chromatography (Step 8) — PCP-008

Process Development / Manufacturing Science & Technology (MSAT)

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Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

AEX, anion exchange chromatography; ALCOA+, attributable, legible, contemporaneous, original, accurate and complete data-integrity principles; AMV, analytical method validation; CCD, central composite design; CEX, cation exchange chromatography;

Cpk, one-sided process capability index; CPP, critical process parameter; CQA, critical quality attribute; DNA, deoxyribonucleic acid; DoE, design of experiments; DS, drug substance; ELISA, enzyme-linked immunosorbent assay; GPP, general process parameter; HCP, host cell protein; icIEF, imaged capillary isoelectric focusing; ICH, International Council for Harmonisation; KPP, key process parameter; LRF, log reduction factor; MSAT, Manufacturing Science and Technology; MVM, minute virus of mice (parvovirus model); NOR, normal operating range; PAR, proven acceptable range; PCP, Process Characterization Plan; PCR, Process Characterization Report; PPQ, process performance qualification; PRESS, prediction sum of squares; qPCR, quantitative polymerase chain reaction; RSM, response-surface methodology; SDM, scale-down model; SOP, standard operating procedure; TCID50, median tissue-culture infective dose; WC-CPP, well-controlled critical process parameter; XMuLV, xenotropic murine leukaemia virus (retrovirus model).

1 Purpose and scope

This plan defines the process characterization study for anion exchange chromatography (AEX), Step 8 of the A-Mab drug substance process, and it fixes the acceptance and decision criteria against which the executed study will be judged. AEX is the final chromatographic step of the purification train and is operated in flow-through mode, so that the antibody passes through the column while acidic impurities and both model viruses bind to the resin. The study will be executed on a qualified scale-down model at the development site before the process is transferred to the commercial drug substance facility (PTP-001).

The step carries the drug substance claim for cumulative parvovirus clearance, and it also clears retrovirus, host cell protein (HCP), residual DNA and leached Protein A. Viral safety is a modular claim built from three steps whose mechanisms are independent of one another, so this plan covers only the contribution made by anion exchange. The contributions of low-pH inactivation and of small-virus retentive filtration are characterized in their own plans (PCP-006 and PCP-009) and are consolidated in PCMR-001.

The plan covers the parameters to be varied and the ranges over which they will be varied, the scale-down model and its qualification (including the small-scale spiking model used for viral clearance), the analytical methods and the sampling plan, the planned statistical analysis, the acceptance and decision criteria, and the deliverables of the study.

Three areas fall outside its scope. Confirmation of the step at commercial scale belongs to process performance qualification and is addressed under the transfer plan (PTP-001), while resin packing, sanitization and lifetime are governed by SOP-2008 and are covered by a separate resin life-cycle study. The clearance credited to the other two viral steps is characterized in their own plans, and this plan neither repeats nor revises it.

2 Objectives

The study has five objectives.

1. Quantify the effect of the anion exchange operating parameters on parvovirus and retrovirus log reduction, on pool host cell protein and on step yield, over ranges wider than those used in routine operation.
2. Define a multivariate operating region over which every governed attribute is predicted to remain within its acceptance criterion.
3. Establish a proven acceptable range (PAR) for each multivariate parameter, both at its set-point and with the remaining parameters varying within their normal operating ranges (NORs).
4. Provide the evidence needed to classify each parameter under SOP-4001, including the evidence needed to classify a parameter that shows no effect.
5. Justify the ranges to be used in Stage 2 manufacture and the contribution this step makes to the drug substance control strategy.

The first objective is the basis for the four that follow, since neither the operating region nor the proven acceptable ranges can be drawn until the parameter effects have been quantified on a model that meets the adequacy criteria of §7.

3 Related documents

The characterization package that frames this study is listed in Table 3.1.

Table 3.1: Related documents in the A-Mab characterization package.

Document ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCR-008	Process Characterization Report (Anion Exchange Chromatography (Step 8))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

The controlled procedures and the validated analytical methods that govern execution of the study are listed in Table 3.2. Each of them is referenced again at the point in this plan where it applies.

Table 3.2: Controlled procedures and validated analytical methods for this step.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2011	Operation of the Anion-Exchange Polishing Chromatography Step	SOP
SOP-2008	Chromatography Resin Life-Cycle, Packing and Sanitization	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3014	Residual DNA (qPCR)	Method validation
AMV-3016	Leached Protein A by ELISA (ppm)	Method validation

Reference	Title	Type
AMV-3017	XMuLV Infectivity Titre (TCID50)	Method validation
AMV-3018	MVM Infectivity Titre (TCID50/qPCR)	Method validation
AMV-3013	Charge Variants (icIEF)	Method validation

4 Prior knowledge and quality risk basis

4.1 Unit-operation description and prior knowledge

Anion exchange chromatography is the final purification step in the A-Mab downstream process, and it uses a strong anion exchange resin operated in flow-through mode. At the load pH the antibody carries a net positive charge and passes through the column, while acidic impurities and both model viruses bind to the resin. The flow-through pool is collected from the start of the load to the end of the first wash (SOP-2011). The column is then stripped, sanitized and stored under the resin life-cycle procedure (SOP-2008).

The step has an established platform history. Flow-through anion exchange has been used at commercial scale for three licensed antibodies produced on the same humanized IgG1 platform (X-Mab, Y-Mab and Z-Mab) as well as for A-Mab clinical supply, and the mechanism of impurity and virus removal is the same in each case (CMC Biotech Working Group 2009). Prior knowledge therefore sets the expected direction of every parameter effect, but it does not remove the need for characterization, because the isoelectric point of A-Mab and the impurity profile of its cation exchange pool both differ from those of the earlier products.

The mechanistic expectation for the study is set out here so that the results reported in PCR-008 can confirm or refute it. Raising the load pH increases the net negative charge carried by host cell protein, by DNA and by the capsid of both model viruses (MVM and XMuLV). Binding to the resin strengthens as a result, and the clearance of all three impurity classes improves. Raising conductivity shields those same charges and weakens binding, so clearance falls. The conductivity of the equilibration and wash-1 buffer governs the ionic environment in which weakly bound impurities are displaced during the wash. The conductivity of the load governs the ionic environment during impurity capture itself. Protein load sets the impurity mass challenge presented to a fixed volume of resin, and it determines how closely the column approaches breakthrough.

Two things are not claimed for this step. Anion exchange operated under platform conditions does not modify the glycan or the charge variant distribution of the antibody, both of which are formed in the production bioreactor and reported in PCR-003. The step is also not claimed to deliver the viral safety of the drug substance on its own.

4.2 Quality attributes in scope

This step sets one quality attribute, and its acceptance criterion is given in Table 4.1. Cumulative parvovirus clearance is of very high criticality under Tool #1 of the A-Mab framework, which scores the impact of an attribute against the uncertainty in that impact (CMC Biotech Working Group 2009). Criticality is treated as a continuum and not as a binary label (ICH Q9), and the attributes of high or very high criticality are those called critical (International Council for Harmonisation 2023b).

Table 4.1: Quality attribute set by this step, with its acceptance criterion.

CQA	Category	Acceptance	Criticality	Tool #1
Viral clearance — MVM (parvovirus)	viral safety	8.6–20 log10 (cumulative)	VH	20

The step also governs several attributes that are formed or set elsewhere (Table 4.2). Retrovirus clearance is set at the low-pH viral inactivation step and is reported in PCR-006. Host cell protein, residual DNA and leached Protein A are formed upstream and are reduced by more than one purification step in the train. This plan measures the contribution that anion exchange makes to each of them, and it claims nothing beyond that contribution.

Table 4.2: Quality attributes governed or cleared by this step.

CQA	Category	Acceptance	Criticality	Tool #1
Viral clearance — MVM (parvovirus)	viral safety	8.6–20 log10 (cumulative)	VH	20
Viral clearance — XMuLV (enveloped)	viral safety	16.7–30 log10 (cumulative)	VH	20
Host Cell Protein (HCP)	process impurity	0–100 ng/mg	M-H	36
Residual DNA	process impurity	0–0.001 ng/dose	VL	6
Leached Protein A	process impurity	0–5 ppm	L-M	16

4.3 Risk-based prioritization of parameters

The scope of this study was set by the pre-characterization risk assessment (RA-001), which ranked every anion exchange parameter on two scales: the potential impact of the parameter on a critical quality attribute, and its potential to interact with other parameters. Where no data or rationale were available to make an assessment, the parameter was ranked at the highest level.

The ranking placed the parameters into three groups. Parameters with a high combined score go to the multivariate design, parameters with a moderate score are assessed one at a time, and parameters with a low score are held at platform conditions and are not studied further, because prior knowledge already bounds them. The resulting scope is 4 parameters in the multivariate design and 1 in the univariate assessment, out of 5 parameters in total (Table 6.1).

The characterization ranges are wider than the normal operating ranges in every case. A wider range resolves a real effect against analytical variability, and it supports later movement within the design space (International Council for Harmonisation 2009). The widening is bounded by what the step can physically deliver. Load pH is bounded above by the point at which the antibody itself begins to bind the resin and below by the buffering capacity of the load, while protein load is bounded above by the dynamic capacity of the resin for the impurities it must retain.

5 Materials and methods

5.1 Scale-down model and its qualification

The study will be run on a laboratory scale-down model (SDM) of the commercial anion exchange column, designed on established principles of chromatography scaling. The model will hold bed height, linear velocity, residence time, protein load per litre of resin and every buffer volume in column volumes at the commercial equivalent, and column efficiency will be verified by plate count and peak asymmetry before each campaign. Qualification will follow SOP-1001.

Qualification will compare the model against at-scale data from A-Mab clinical manufacture, using the input and output attributes of the step (pool HCP, residual DNA, leached Protein A and step yield). The model will be accepted for use in characterization when no statistically significant difference between scales is found for any of those attributes. The qualification record will be filed with the study data and summarized in PCR-008.

Viral clearance will be studied in a separate small-scale spiking model held in the containment laboratory, as required by ICH Q5A (International Council for Harmonisation 2023a). The spiking model will use the same column geometry, the same buffers and the same residence time as the non-spiked model, and spiked material will not be returned to the process. The spike will be added to the load and the load will then be filtered before application, so that spike-derived particulates do not alter the behaviour of the column. Virus titre will be measured in the load and in the flow-through pool. The detection limit of each infectivity assay, together with any cytotoxic or interfering effect of the matrix, will be established before the study begins and reported in PCR-008.

The centre point replicates carry a second role, since they provide the pure error estimate against which lack of fit is tested. 3 replicates are planned in the screening design and 4 in the response-surface design.

5.2 Operation

The step will be operated as described in SOP-2011. The column will first be equilibrated in the equilibration and wash-1 buffer at the target conductivity. The cation exchange pool will then be titrated to the target load pH, adjusted to the target conductivity and applied to the column. The flow-through will be collected from the start of the load to the end of the first wash, after which the column will be stripped, sanitized and stored under SOP-2008.

Resin lot, buffer composition and column packing are controlled to platform specification, so they are identity and quality controlled inputs and are not characterized in this study. Resin cycle number will be recorded for every run, and runs will be distributed across cycles so that no corner of the design is confounded with resin age.

5.3 Analytical methods

The validated methods to be used are listed in Table 3.2. Pool host cell protein will be measured by ELISA (AMV-3012), residual DNA by qPCR (AMV-3014) and leached Protein A by ELISA (AMV-3016). Retrovirus and parvovirus titres will be measured by infectivity assay in the containment laboratory (AMV-3017 and AMV-3018). Acidic charge variants in the load will be measured by icIEF (AMV-3013), which is how each load lot will be confirmed representative before it is used (see §12). Method performance is held in the validation records and is not repeated in this plan.

5.4 Sampling plan

Every design point will be executed as a pair of runs: a non-spiked run that provides pool host cell protein, residual DNA, leached Protein A and step yield, and a virus-spiked run on the same load and the same column that provides the two log reduction values. The paired runs will use the same buffer lots.

Samples will be taken from the load, from the flow-through pool and from the strip. The load sample fixes the input attribute values for that run, the pool sample carries the reported responses, and the strip sample supports the impurity mass balance. Every sample will be given a unique identifier at the point of collection, and it will be stored under the conditions given in the relevant method validation.

5.5 Statistical methods

All analysis will follow SOP-1002. Factors will be expressed in coded units, where the low edge of the characterization range is -1, its midpoint is 0 and its high edge is $+1$. All effects will be estimated on that coded scale, so that they are comparable across parameters. An effect is reported as twice the coded coefficient, so it is the change in the response across half the range studied.

The screening data will be fitted by ordinary least squares with all main effects and all two-factor interactions, a model of 11 terms that leaves 8 residual degrees of freedom. The response-surface data will be fitted with a full quadratic model of 15 terms, which leaves 13 residual degrees of freedom. Every term will be reported with its standard error and its p-value, and significance will be judged at $p < 0.05$.

Model adequacy will be judged on the coefficient of determination, the adjusted coefficient, the predicted coefficient computed from the prediction sum of squares

(PRESS), and the overall F test. Lack of fit will be tested against the pure error from the centre replicates. The residual diagnostics will be inspected for structure against the predicted value and for departure from normality.

Commercial-scale capability will be estimated by Monte-Carlo simulation of 2,000 batches, drawn with every parameter varying within its normal operating range. A one-sided capability index (Cpk) will then be computed for each governed attribute against its acceptance criterion.

6 Study design

6.1 Factors, ranges and study type

The design resolves the parameter effects in two stages. A screening design will identify which parameters affect the governed responses, and a response-surface design will then model those responses across the region. The response-surface model is the predictive model and the basis of the operating region, whereas the screening fit is near-saturated and will not be used for that purpose.

The parameters to be studied, their set-points, their characterization ranges and their normal operating ranges are given in Table 6.1, and the coded letters used in the design matrices are given in Table 6.2 together with the natural value at each coded level and the ratio of the characterization range to the normal operating range.

Table 6.1: Parameters to be studied, with set-points, characterization ranges, normal operating ranges and study type.

Parameter	Unit	Set-point	Range studied	NOR	Study type
Load pH	pH	7.5	7.2–7.8	7.35–7.65	multivariate
Equil/Wash-1 conductivity	mS/cm	2.6	1.6–3.6	2.1–3.1	multivariate
Load conductivity	mS/cm	5.5	3–8	4–7	multivariate
Protein load	g/L resin	200	50–300	120–280	multivariate
Operating flow rate	cm/hr	300	150–450	200–400	univariate

Table 6.2: Coded factor legend for the multivariate designs.

Code	Factor	Unit	Low (−1)	Centre (0)	High (+1)	Set-point	NOR	Range / NOR
A	Load pH	pH	7.2	7.5	7.8	7.5	7.35–7.65	2
B	Equil/Wash-1 conductivity	mS/cm	1.6	2.6	3.6	2.6	2.1–3.1	2
C	Load conductivity	mS/cm	3	5.5	8	5.5	4–7	1.67

Code	Factor	Unit	Low (−1)	Centre (0)	High (+1)	Set-point	NOR	Range / NOR
D	Protein load	g/L resin	50	175	300	200	120–280	1.56

The centre point of the design is the midpoint of each characterization range. For load pH, for equilibration and wash-1 conductivity and for load conductivity that midpoint coincides with the commercial set-point, but for protein load it lies below the set-point, because the range was extended further downward than upward in order to cover the low-load condition of a short campaign batch. The set-point of every factor is given in Table 6.2, so the offset is visible.

Each range is justified separately. Load pH will be studied across the range over which the neutralized cation exchange pool can be held without approaching the isoelectric point of the antibody, and the conductivity of the equilibration and wash-1 buffer will be studied across the range that buffer preparation can deliver reproducibly. Load conductivity will be studied up to the highest value a neutralized cation exchange pool can reach, so that the study bounds the worst input this step can receive. Protein load will be studied from a low value typical of a short campaign batch up to a value above the highest planned commercial load, and operating flow rate will be studied across the range the commercial skid can deliver.

6.2 Screening design

The screening design is a two-level full factorial in the 4 multivariate factors augmented with 3 centre points, giving 19 runs in total. A full factorial was chosen over a fractional design, because 4 factors are few enough for the full design to be affordable at scale-down. It estimates all 4 main effects and all 6 two-factor interactions without aliasing. Runs will be randomized within the campaign, and the planned coded matrix is given in Appendix A.

The screening stage identifies the factors that matter. It does not define the operating region.

6.3 Response-surface design

The response-surface design is a face-centred central composite design (CCD) over the same 4 factors, comprising 16 factorial runs, 8 axial runs and 4 centre points, or 28 runs in total. The axial runs sit on the faces of the cube, so that no run falls outside the characterization range and every setting stays inside the range the step can actually deliver. That choice costs some precision on the pure quadratic terms, and it is stated here because the operating region will rest on this model.

All 4 multivariate factors are carried into the response-surface design, since each is expected to act on at least one governed response by the mechanism given in §4.1. A factor found inactive for every governed response at screening will be dropped from the fitted model. It will be retained in the knowledge space as evidence of robustness, and its classification will be argued from that null result. The planned coded matrix is given in Appendix B.

6.4 Univariate assessment

Operating flow rate will be assessed one factor at a time, at the low edge, the centre and the high edge of its characterization range, with every other parameter held at its set-point. Flow rate acts on this step through residence time and not through the binding chemistry, so an interaction with pH or with conductivity is not expected. On that basis a univariate assessment over three levels is sufficient to bound its range. If the assessment shows an effect on any governed response, the parameter will be added to a supplementary multivariate design and PCR-008 will report that change.

Residual DNA and leached Protein A will be measured across both multivariate designs but are not carried as response-surface responses. Platform experience indicates that this step clears both to well below their drug substance limits (Table 4.2), and the study will confirm that clearance without modelling it.

7 Acceptance and decision criteria

The study will be judged against criteria fixed before execution, and they fall into three groups: the acceptance criterion for each response, the adequacy criterion for each fitted model, and the rules by which the operating region and the proven acceptable ranges will be declared.

The acceptance criterion for each response is given in Table 7.1. Pool host cell protein is judged against an in-process limit derived from the drug substance specification of 100 ng/mg by dividing it by an assurance margin. The step must meet the drug substance limit at its own outlet, because no later step in the train is credited with host cell protein clearance, and the margin is what keeps a batch at the in-process limit clear of the specification. Step yield carries no acceptance criterion and will be reported as a process performance attribute.

Table 7.1: Acceptance criteria for the study responses.

Response	Unit	Acceptance basis	Criterion
Pool HCP (ng/mg)	ng/mg	Drug-substance specification	≤ 100
XMuLV LRF ($\log_{10}$)	$\log_{10}$ per step	Back-calculated step contribution	≥ 3.78
MVM LRF ($\log_{10}$)	$\log_{10}$ per step	Back-calculated step contribution	≥ 3.28
Step yield	fraction of load	Process performance	Reported, no acceptance criterion

The two viral criteria are derived differently, because each is a step contribution to a cumulative requirement, so the criterion for this step is the cumulative requirement minus the clearance credited to the other steps (Table 7.2). For parvovirus the cumulative requirement is 8.6 $\log_{10}$ and the other steps are credited with 5.32 $\log_{10}$, so the requirement at this step is 3.28 $\log_{10}$. The corresponding requirement for retrovirus is 3.78 $\log_{10}$.

Table 7.2: Back-calculation of the step-level viral clearance criteria ($\log_{10}$).

Model virus	Cumulative requirement	Credited to the other steps	Required at this step
XMuLV	16.70	12.92	3.78
MVM	8.60	5.32	3.28

Low-pH inactivation contributes no clearance for a non-enveloped parvovirus, so the cumulative parvovirus claim rests on this step and on virus filtration alone, which is why anion exchange is the step that sets the attribute. The credited contributions of the other steps will be re-checked against the consolidated figures in PCMR-001 before PCR-008 is issued.

A model will be accepted as adequate when the overall F test is significant at $p < 0.05$. In addition, the lack-of-fit F test must not be significant at the same level when it is tested against the pure error from the centre replicates. The residual diagnostics must show no structure against the predicted value and no material departure from normality. The predicted coefficient of determination must be positive, and it will be reported alongside the adjusted coefficient for every response.

Only a model that meets all of those conditions will be used to define the operating region. A model that fails the lack-of-fit test will be re-specified once against the same data set. If it fails again, the operating region for that response will be set from the univariate data and PCR-008 will state the limitation.

The operating region will be declared as the multivariate set of parameter settings over which every governed response is predicted by the response-surface model to stay within its acceptance criterion. It will be reported with the normal operating range located inside it and the characterization range drawn around it, and the worst-case corner of the region will be identified by name. Movement within the declared region is not a change to the process, whereas movement outside it requires change control under the pharmaceutical quality system (International Council for Harmonisation 2008).

The study passes when three conditions hold: every governed response meets its acceptance criterion at the set-point and across the normal operating range, every multivariate parameter has a proven acceptable range that contains its normal operating range, and every parameter has a classification supported by the data under SOP-4001.

The study fails, in whole or in part, if a governed response breaches its acceptance criterion anywhere inside the normal operating range. In that event the normal operating range will be narrowed to the proven acceptable range, or the parameter will be reclassified as a critical process parameter, or both, and any such outcome will be recorded in PCR-008 with the analysis that supports it.

Capability will be assessed against the minimum index defined for the campaign in PCMP-001. Capability summarizes the operating region and is not an independent criterion, so a shortfall will be resolved by narrowing the normal operating range and not by relaxing an acceptance criterion.

Any departure from this plan will be recorded as a deviation, investigated and impact-assessed in PCR-008. A deviation that renders a data set unrepresentative will invalidate the affected design, which will then be re-executed in full on requalified material.

8 Proven acceptable ranges (planned analysis)

A proven acceptable range will be determined for every combination of governed attribute and multivariate parameter, on the acceptance basis given in §7: the drug substance specification for pool host cell protein, and the back-calculated step contribution for the two viral responses.

Two analyses will be run for each combination. The first holds the other parameters at their set-points and evaluates the fitted response-surface model across the parameter's characterization range. The second varies the other parameters within their normal operating ranges by Monte-Carlo simulation of the same model. The propagated analysis uses 2,000 draws at each of 81 grid points across the range, and it adds the residual variation of the model to every draw.

The decision rule is the same in both analyses: the proven acceptable range is the contiguous interval of the parameter, containing the set-point, over which the attribute stays within acceptance. In the analysis at set-point the predicted mean must stay within acceptance, whereas in the propagated analysis the 95 per cent predictive interval must stay within acceptance, which is a robustness criterion and gives the narrower of the two ranges. Where the set-point itself breaches acceptance, no proven acceptable range exists for that combination and the report will say so.

The planned analysis will be reported in one table and one plot for each governed attribute. The table will give, for every attribute and parameter, the characterization range and both proven acceptable ranges. The plot will carry the parameter on the horizontal axis and the attribute on the vertical axis, with the acceptance limit drawn, the acceptable region shaded green, and the set-point and the normal operating range marked. The two analyses will appear as the two panels of the plot.

The relationship between these ranges and the operating region will be stated in the report, since a proven acceptable range is a univariate statement about one parameter while the operating region is a multivariate statement about all of them together. Where the whole characterization range of a parameter is proven acceptable, the binding constraint on the operating region is the multivariate corner and not that parameter's univariate range. Every conclusion drawn from this analysis will be bounded by the characterization ranges and by the fitted model, and neither extends beyond the settings actually run.

9 Data management and integrity

All study data will be recorded in accordance with ALCOA+ principles. Raw analytical data will be captured in the validated laboratory information system and run records in the electronic batch record of the development laboratory, and the statistical analysis will be performed on a locked data extract that is archived together with the analysis script and its output.

Every record will be reviewed by a second qualified person before the data are used, and a change to a recorded value will be made under audit trail with the reason recorded. The analysis is reproducible from the archived extract and script, and PCR-008 will name the archive location.

10 Roles and responsibilities

Responsibilities are assigned as shown in Table 10.1. Process Development owns the plan, its execution and the report, while Quality Assurance approves both documents and dispositions any deviation.

Table 10.1: Roles and responsibilities for the study.

Function	Responsibility
Process Development / MSAT	Authors the plan, executes the studies, fits the models, authors PCR-008
Analytical Development	Provides and maintains the validated methods; releases analytical data
Virology laboratory	Executes the spiked runs and the infectivity assays under containment
Manufacturing Science	Confirms commercial equivalence of the scale-down model and the operating ranges
Statistics	Approves the designs, the analysis plan and the fitted models
Quality Assurance	Approves the plan and the report; dispositions deviations

11 Deliverables and schedule

The deliverable of this study is PCR-008, the process characterization report for this step, which will present the executed designs, the fitted models, the operating region, the proven acceptable ranges, the capability estimate and the parameter classifications. It will also record any deviation from this plan together with the impact assessment of that deviation.

The study will run in four phases. Scale-down model qualification comes first and includes qualification of the small-scale spiking model, the screening design follows and is analysed before the response-surface design is executed, and the response-surface design and the univariate assessment then run together. The report is written last. No calendar dates are fixed in this plan, and the campaign schedule is held in PCMP-001.

12 Risks and assumptions

The principal risk to this study is an unrepresentative load. The load for every run is a cation exchange pool whose charge variant content depends on how long it is held after neutralization, and an elevated acidic charge variant burden alters the charge balance the column sees and can change the apparent parameter effects. For this reason the acidic charge variant content of every load lot will be measured by icIEF (AMV-3013) before use, a lot outside the platform range will be rejected, and any run already performed on such a lot will be repeated on requalified material.

The second risk is the representativeness of the scale-down model, which is qualified on input and output attributes at the set-point and whose qualification does not extend to the edges of the characterization ranges. This limitation will be stated in PCR-008, and it is managed by confirming the step at commercial scale during process performance qualification (U.S. Food and Drug Administration 2011).

The third risk sits in the small-scale spiking model, because spike preparations can carry cytotoxic or interfering material and every infectivity assay has a finite detection limit. A log reduction bounded by the detection limit will be reported as a greater-than value and will not be extrapolated, and both effects will be quantified during assay qualification and reported in PCR-008.

The fourth risk is material supply, since the study needs one resin lot and consistent buffer lots across the campaign. If a lot changes during the campaign, the change will be recorded and the affected runs will be identified in the report.

Three assumptions underlie the plan. Prior platform knowledge is assumed to give the correct direction of each parameter effect, and the study is designed to test that assumption and not to rely on it. The clearance credited to low-pH inactivation and to virus filtration is assumed to hold at the values recorded in the current viral safety assessment, and the analytical methods are assumed to remain in a validated state for the duration of the campaign.

13 Approvals

This plan becomes effective when it has been approved by the functions listed in Table 1, and execution may not begin before that approval is complete. Any change to the plan after approval will be made by revising the whole document under the same approval route.

14 Appendices A–B — Planned design matrices

The planned coded design matrices are given below, without response columns, because no data exist at the time this plan is issued. Run order will be randomized at execution, so the run numbers given are design identifiers only.

14.1 Appendix A — Planned screening design matrix

The coded settings of the 19 planned screening runs are given in Table 14.1, and the factor letters are defined in Table 6.2.

Table 14.1: Planned coded screening design matrix (two-level full factorial with centre points).

Run	Type	A	B	C	D
1	factorial	-1	-1	-1	-1
2	factorial	-1	-1	-1	1
3	factorial	-1	-1	1	-1
4	factorial	-1	-1	1	1
5	factorial	-1	1	-1	-1
6	factorial	-1	1	-1	1
7	factorial	-1	1	1	-1
8	factorial	-1	1	1	1
9	factorial	1	-1	-1	-1
10	factorial	1	-1	-1	1
11	factorial	1	-1	1	-1
12	factorial	1	-1	1	1
13	factorial	1	1	-1	-1
14	factorial	1	1	-1	1
15	factorial	1	1	1	-1
16	factorial	1	1	1	1
17	center	0	0	0	0
18	center	0	0	0	0
19	center	0	0	0	0

14.2 Appendix B — Planned response-surface design matrix

The coded settings of the 28 planned response-surface runs are given in Table 14.2.

Table 14.2: Planned coded response-surface design matrix (face-centred central composite).

Run	Type	A	B	C	D
1	factorial	-1	-1	-1	-1
2	factorial	-1	-1	-1	1
3	factorial	-1	-1	1	-1
4	factorial	-1	-1	1	1
5	factorial	-1	1	-1	-1
6	factorial	-1	1	-1	1
7	factorial	-1	1	1	-1
8	factorial	-1	1	1	1
9	factorial	1	-1	-1	-1
10	factorial	1	-1	-1	1
11	factorial	1	-1	1	-1
12	factorial	1	-1	1	1
13	factorial	1	1	-1	-1
14	factorial	1	1	-1	1
15	factorial	1	1	1	-1
16	factorial	1	1	1	1
17	axial	-1	0	0	0
18	axial	1	0	0	0
19	axial	0	-1	0	0
20	axial	0	1	0	0
21	axial	0	0	-1	0
22	axial	0	0	1	0
23	axial	0	0	0	-1
24	axial	0	0	0	1
25	center	0	0	0	0
26	center	0	0	0	0
27	center	0	0	0	0
28	center	0	0	0	0

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